Agnostic Life Finder

to search for extant life in Martian Ice

Špaček, Jan[1]; Benner, Steven[1]; Temby, Chris[1]; Alpern, Holden[1,2]; Zeffren, Joshua[1,2]; George, Alaysia[1,2]; Montaudon, Eliott[1,2]; Rizzo, Gabriella[3]; Young, Izabelle[1,2]; Alisha, Hunnewell[1,2]; Fuchs, Lia[1,2]

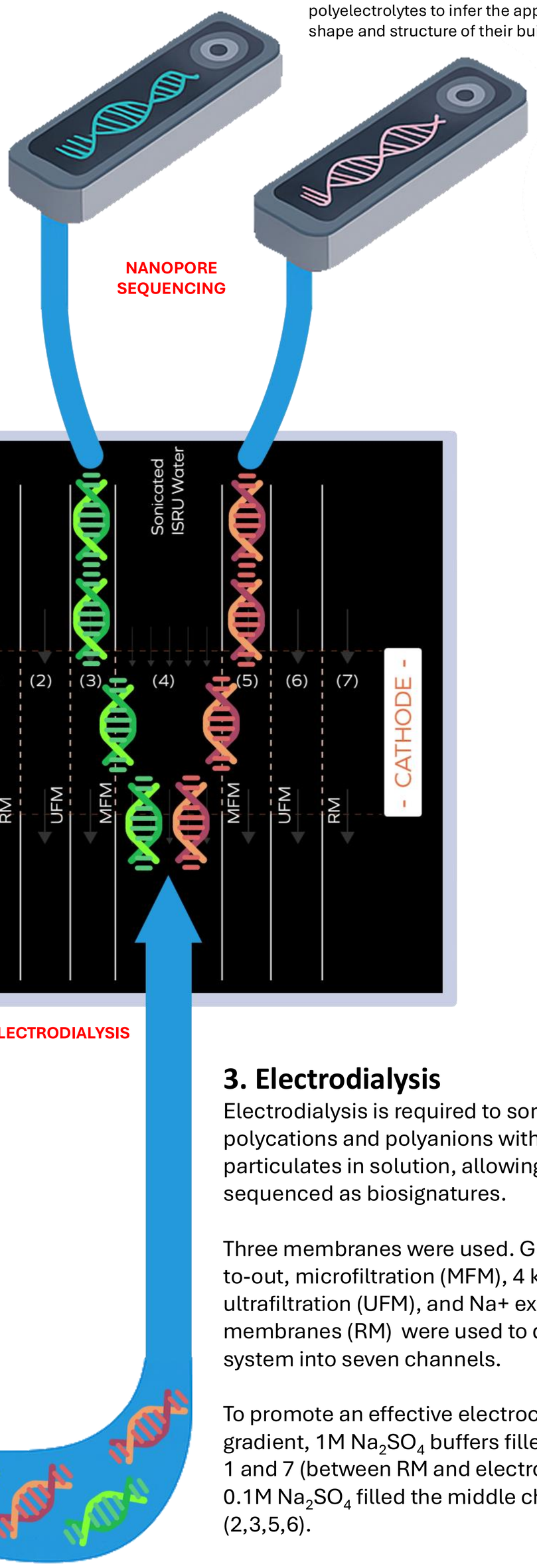
Introduction

The Agnostic Life Finder (ALF) is a laboratory-tested concept for determining if Martian ice harbors life. It relies on the polyelectrolyte theory of the gene, proposes that any system capable of Darwinian evolution requires a charged polyelectrolyte like DNA to store genetic information.

After sonication lyses potential cells, ALF uses electrodialysis to separate polyelectrolytes from ordinary molecules via their charge. The polyelectrolytes are trapped in the third and fifth channels due to porous membranes and then sequenced with Nanopore technology.

ALF provides a novel, now tested method to find agnostic biosignatures on worlds with extractable water.

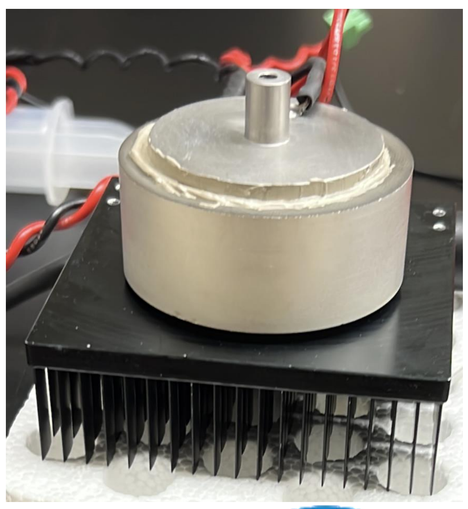
When charged polyelectrolytes pass through nanopores, distinct electrical signals are produced, depending on the details of the molecule. Solid-state nanopores offer single-molecule detection and can analyze linear polyelectrolytes to infer the approximate shape and structure of their building blocks.





Final sonicator design on top of heatsink (right)

Full sonicator system (left)
A: Ultrasonic Driver
B: Syringe Pump
C: Heatsink
D: Transducer
E: Fan
F: Software



1. Extraction

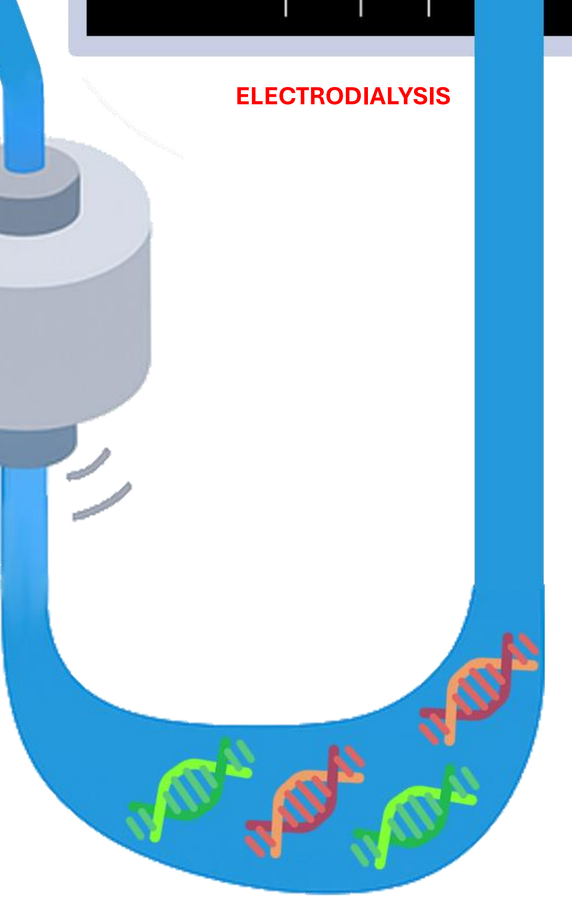
ALF would take advantage of existing plans to mine Martian ice for electrolysis, which would provide oxygen gas for in-situ rocket fuel production. One method of extraction is the "Rodwell," which reheats extracted water to melt more ice, allowing for a continuous flow into ALF. A model is already in-development by Honeybee Robotics.

2. Sonication

Any cell-like structures from the extracted groundwater must be lysed. While other methods were considered, the cylindrical, flow-through sonicator shown below was chosen due to its simplicity.

- Mechanical lysis could damage system
- Chemical lysis requires large buffer quantities

We demonstrated that our sonicator can disrupt cells like *E. coli,* including hard-to-sonicate yeasts (*S. cerevisiae*). Sonicated *E. coli* cells were run through an ALF prototype, where an estimated 41% of DNA was collected in channel 3 (anode side).



Gels were from sample of DNA and *E. coli* run through a flow-through sonicator prototype.

Note that the sonicator was able to lyse *E. coli* (last three rows) and shear a significant amount (but not all) of its DNA to under 500 bp.

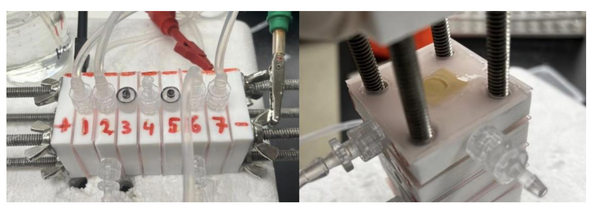
3. Electrodialysis

Electrodialysis is required to sort polycations and polyanions with other particulates in solution, allowing them to be sequenced as biosignatures.

Three membranes were used. Going from in-to-out, microfiltration (MFM), 4 kDa ultrafiltration (UFM), and Na+ exchange membranes (RM) were used to divide the system into seven channels.

To promote an effective electrochemical gradient, 1M Na_2SO_4 buffers filled channels 1 and 7 (between RM and electrodes), while 0.1M Na_2SO_4 filled the middle channels (2,3,5,6).

The system successfully moved both polyanions and polycations alike to channels 3 and 5, respectively. 75% of DNA was extracted during an overnight run, concentrating the DNA from an initial 317.3 ng/μL to an average of 903.4 ng/μL (measured via Nanodrop One).



Preliminary "small ALF" model used with static samples. Note the labelled cathode, anode, and channels in the picture on the left. The right picture shows the Na+ exchange membrane covering the hole between channels 1 and 2.